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Foreword

This European Telecommunication Standard (ETS) has been produced by the Digital Enhanced Cordless Telecommunications (DECT) Project of the European Telecommunications Standards Institute (ETSI).

Transposition dates	
Date of adoption:	23 May 1997
Date of latest announcement of this ETS (doa):	30 September 1997
Date of latest publication of new National Standard or endorsement of this ETS (dop/e):	31 March 1998
Date of withdrawal of any conflicting National Standard (dow):	31 March 1998

Introduction

To evaluate conformance of a particular implementation, it is necessary to have a statement of which capabilities and options have been implemented for a telecommunication specification. Such a statement is called an Implementation Conformance Statement (ICS).

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1 Scope

This European Telecommunication Standard (ETS) provides the Implementation Conformance Statement (ICS) proforma for the Digital Enhanced Cordless Telecommunications (DECT) Authentication Module (DAM) and portable equipment defined in ETS 300 331 [1] in compliance with the relevant requirements, and in accordance with the relevant guidance given in ISO/IEC 9646-7 [4] and ETS 300 406 [2].

2 Normative references

This ETS incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] ETS 300 331 (1995): "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); DECT Authentication Module (DAM)".

[2] ETS 300 406 (1995): "Methods for Testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

[3] ISO/IEC 9646-1 (1995): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[4] ISO/IEC 9646-7 (1995): "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of this ETS, the following definitions apply:

- terms defined in ETS 300 331 [1];
- terms defined in ISO/IEC 9646-1 [3] and in ISO/IEC 9646-7 [4].

In particular, the following terms defined in ISO/IEC 9646-1 [3] apply:

Implementation Conformance Statement (ICS): A statement made by the supplier of an implementation or system claimed to conform to a given specification, stating which capabilities have been implemented. The ICS can take several forms: protocol ICS, profile ICS, profile specific ICS, information object ICS, etc.

ICS proforma: A document, in the form of a questionnaire, which when completed for an implementation or system becomes an ICS.

3.2 Symbols

For the purposes of this ETS, the following symbols apply:

{} Optional data, e.g. "CLA, INS, P1, P2, P3 {, data}" indicates that data may or may not follow the CLA, INS, P1, P2, P3 bytes.

3.3 Abbreviations

For the purposes of this ETS, the following abbreviations apply:

DECT	Digital Enhanced Cordless Telecommunications
ICS	Implementation Conformance Statement
IUT	Implementation Under Test
PE	Portable Equipment
SCS	System Conformance Statement
SUT	System Under Test
UAK	User Authentication Key

4 Conformance to this ICS proforma specification

If it claims to conform to this ETS, the actual ICS proforma to be filled in by a supplier shall be technically equivalent to the text of the ICS proforma given in annex A, and shall preserve the numbering/naming and ordering of the proforma items.

An ICS which conforms to this ETS shall be a conforming ICS proforma completed in accordance with the guidance for completion given in clause A.1.

Annex A (normative): ICS proforma for ETS 300 331

Notwithstanding the provisions of the copyright clause related to the text of this ETS, ETSI grants that users of this ETS may freely reproduce the ICS proforma in this annex so that it can be used for its intended purposes and may further publish the completed ICS.

A.1 Guidance for completing the ICS proforma

A.1.1 Purposes and structure

The purpose of this ICS proforma is to provide a mechanism whereby a supplier of an implementation of the requirements defined in ETS 300 331 [1] may provide information about the implementation in a standardized manner.

The ICS proforma is subdivided into subclauses for the following categories of information:

- guidance for completing the ICS proforma;
- identification of the implementation;
- identification of the standard;
- global statement of conformance;
- roles;
- DECT Authentication Module (DAM):
    - physical characteristics:
        - format and layout:
            - ID-1 card;
            - plug-in card;
        - Contacts;
    - electronic signals and transmission protocols:
        - supply Voltage Vcc (contact C1);
        - Reset RST (contact C2);
        - Clock CLK (contact C3);
        - I/O (contact C7);
        - states;
        - Answer To Reset (ATR):
            - ATR: TC1 parameter values;
    - logical model:
        - file identifier;
        - dedicated files;
        - elementary files;
        - methods for selecting the DECT application;
        - reservation of file IDs:
            - DFs;
            - EFs;
    - security services and facilities:
        - algorithms and processes;
        - authentication;
        - UAK allocation;
        - file access control;
    - description of the functions;
    - description of the commands:
        - mapping principles;
        - coding of the commands;
        - definitions and coding;
        - status conditions returned by the DAM:
            - coding of the status words;
            - commands versus possible status responses;
    - contents of the elementary files:
        - contents of the EFs at the MF level:
            - optional data parameters in EF_{ICC};
            - optional data parameters in EF_{ID};
            - optional data parameters in EF_{NAME};
            - optional data parameters in EF_{DIR};

        - optional data parameters in EF_{LANG};
    - contents of the EFs at the parent level of the DECT application;
    - contents of the EFs at the DECT application level:
        - optional data parameters in EF_{LSR};
    - contents of the EFs at the subscription registration level:
        - DF_{SR1}:
            - optional data parameters in EF_{PARK};
            - optional data parameters in EF_{UAK};
            - optional data parameters in EF_{AC};
        - DF_{SR2}:
            - optional data parameters in EF_{PARK};
            - optional data parameters in EF_{UAK};
            - optional data parameters in EF_{AC};
- DECT Portable Equipment (PE):
    - physical characteristics;
    - electronic signals and transmission protocols:
        - supply Voltage Vcc (contact C1);
        - Reset RST (contact C2);
        - programming Voltage Vpp (contact C6);
        - Clock CLK (contact C3);
        - I/O (contact C7);
        - states;
        - Answer To Reset (ATR);
    - logical model:
        - methods for selecting the DECT application;
    - description of the commands:
        - mapping principles;
        - coding of the commands;
    - application protocol:
        - general procedures;
        - DAM management procedures;
        - CHV related procedures;
        - authentication procedures;
        - UAK allocation;
        - general information procedures;
        - subscription registration maintenance.

A.1.2 Abbreviations and conventions

The ICS proforma contained in this annex is comprised of information in tabular form in accordance with the guidelines presented in ISO/IEC 9646-7 [4].

Item column

The item column contains a number which identifies the item in the table.

Item description column

The item description column describes in free text each respective item (e.g. parameters, timers, etc.). It implicitly means "is <item description> supported by the implementation?".

Status column

The following notations, defined in ISO/IEC 9646-7 [4], are used for the status column:

m	mandatory - the capability is required to be supported.
o	optional - the capability may be supported or not.
n/a	not applicable - in the given context, it is impossible to use the capability.
x	prohibited (excluded) - there is a requirement not to use this capability in the given context.
o.i	qualified optional - for mutually exclusive or selectable options from a set. "i" is an integer which identifies an unique group of related optional items and the logic of their selection which is defined immediately following the table.
ci	conditional - the requirement on the capability ("m", "o", "x" or "n/a") depends on the support of other optional or conditional items. "i" is an integer identifying an unique conditional status expression which is defined immediately following the table.

Reference column

The reference column gives reference to ETS 300 331 [1], except where explicitly stated otherwise.

Support column

The support column shall be filled in by the supplier of the implementation. The following common notations, defined in ISO/IEC 9646-7 [4], are used for the support column:

Y or y	supported by the implementation;
N or n	not supported by the implementation;
N/A, n/a or -	no answer required (allowed only if the status is n/a, directly or after evaluation of a conditional status).

If this ICS proforma is completed in order to describe a multiple-profile support in a system, it is necessary to be able to answer that a capability is supported for one profile and not supported for another. In that case, the supplier shall enter the unique reference to a conditional expression, preceded by "?" (e.g. ?3). This expression shall be given in the space for comments provided at the bottom of the table. It uses predicates defined in the SCS, each of which refers to a single profile and which takes the value TRUE if and only if that profile is to be used.

EXAMPLE 1: ?3: IF prof1 THEN Y ELSE N

It is also possible to provide a comment to an answer in the space provided at the bottom of the table.

References to items

For each possible item answer (answer in the support column) within the ICS proforma exists a unique reference, used, for example, in the conditional expressions. It is defined as the table identifier, followed by a solidus character "/", followed by the item number in the table. If there is more than one support column in a table, the columns shall be discriminated by letters (a, b, etc.), respectively.

EXAMPLE 2: A.5/4 is the reference to the answer of item 4 in table A.5 of annex A.

EXAMPLE 3: A.6/3b is the reference to the second answer (i.e. in the second support column) of item 3 in table A.6 of annex A.

Prerequisite line

A prerequisite line takes the form: Prerequisite: <predicate>.

A prerequisite line after a clause or table title indicates that the whole clause or the whole table is not required to be completed if the predicate is FALSE.

A.1.3 Instructions for completing the ICS proforma

The supplier of the implementation shall complete the ICS proforma in each of the spaces provided. In particular, an explicit answer shall be entered, in each of the support boxes provided, using the notation described in subclause A.1.2.

However, the tables containing in the "DECT Authentication Module" subclause shall only be completed for DAM implementations, and the tables containing in the "DECT Portable Equipment" subclause shall only be completed for PE implementations.

If necessary, the supplier may provide additional comments in space at the bottom of the tables, or separately on sheets of paper.

More detailed instructions are given at the beginning of the different subclauses of the ICS proforma.

A.2 Identification of the implementation

Identification of the Implementation Under Test (IUT) and the system in which it resides (the System Under Test (SUT)) should be filled in so as to provide as much detail as possible regarding version numbers and configuration options.

The product supplier information and client information should both be filled in if they are different.

A person who can answer queries regarding information supplied in the ICS should be named as the contact person.

A.2.1 Date of the statement

.....

A.2.2 Implementation Under Test (IUT) identification

IUT name:

.....

.....

IUT version:

.....

A.2.3 System Under Test (SUT) identification

SUT name:

.....

.....

Hardware configuration:

.....

.....

.....

Operating system:

.....

A.2.4 Product supplier

Name:

.....

Address:

.....

.....

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

.....

A.2.5 Client (if different from product supplier)

Name:

.....

Address:

.....

.....

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

.....

A.2.6 ICS contact person

(A person to contact if there are any queries concerning the content of the ICS).

Name:

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

A.3 Identification of the standard

This ICS proforma applies to the following standard:

ETS 300 331 (1995): "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); DECT Authentication Module".

A.4 Global statement of conformance

Are all mandatory capabilities implemented? (Yes/No) .....

NOTE: Answering "No" to this question indicates non-conformance to the standard specification. Non-supported mandatory capabilities are to be identified in the ICS, with an explanation of why the implementation is non-conforming, on pages attached to the ICS proforma.

A.5 Roles

This clause allows the supplier of the implementation to record whether they have implemented a DECT Authentication Module (DAM) or DECT Portable Equipment (PE) from the standard specification. This governs whether the supplier of the implementation needs to fill in the ICS proforma tables in clause A.6 or clause A.7 of this annex.

Table A.1: Roles

Item	Role	Reference	Status	Support
1	DECT Authentication Module (DAM)	1	o.1	
2	DECT Portable Equipment (PE)	1	o.1	

o.1: It is mandatory to support exactly one of these items.

Comments:

A.6 DECT Authentication Module (DAM)

This clause contains the ICS proforma tables related to the DAM role. They are needed to be completed only for DAM implementations:

Prerequisite: A.1/1 -- DAM implementation

A.6.1 Physical characteristics

Table A.2: Major capabilities

Item	Physical characteristic	Reference	Status	Support
1	Physical characteristics in accordance with ISO 7816-1 and ISO 7816-2 unless otherwise specified	4	m	
2	ID-1 card	4.1.1	o.2	
3	Plug-in card	4.1.2	o.2	
4	Temperature range -25°C to +70°C with occasional peaks up to +85°C	4.2	m	
5	Contact pressure ≤ 0,5N per contact	4.3.4	m	

o.2: It is mandatory to support at least one of these items.

Comments:

A.6.1.1 Format and layout

A.6.1.1.1 ID-1 card

Table A.3: ID-1 card

Prerequisite: A.2/2 -- ID-1 card

Item	Physical characteristic	Reference	Status	Support
1	Identification number on the card	4.1	m	
2	Format and layout in accordance with ISO 7816-1 and ISO 7816-2	4.1.1	m	
3	Polarization mark	4.1.1	m	
4	Embossing	4.1.1	o	
5	Embossing in accordance with ISO 7811-1 and ISO 7811-3	4.1.1	c301	
6	Contacts located on the front of the card	4.1.1	o.3	
7	Contacts located on the back of the card	4.1.1	o.3	

c301: IF A.3/4 THEN m ELSE n/a -- ID-1 card is embossed
o.3: It is mandatory to support exactly one of these items.

Comments:

A.6.1.1.2 Plug-in card

Table A.4: Plug-in card

Prerequisite: A.2/3 -- Plug-in card

Item	Physical characteristic	Reference	Status	Support
1	Individual account identifier and check digit	4.1	m	
2	Width = 25 mm	4.1.2	m	
3	Height = 15 mm	4.1.2	m	
4	Thickness the same as an ID-1 type	4.1.2	m	
5	Feature for orientation	4.1.2	m	

Comments:

A.6.1.2 Contacts

Table A.5: Provision of contacts

Item	Contacts	Reference	Status	Support
1	Contact C4 provided	4.3.1	o	
2	Contact C6 bonded only to supply Vpp	4.3.1	m	
3	Contact C8 provided	4.3.1	o	

Comments:

A.6.2 Electronic signals and transmission protocols

Table A.6: Major capabilities

Item	Capability	Reference	Status	Support
1	Electronic signals and transmission protocols in accordance with ISO/IEC 7816-3, unless otherwise specified	5	m	
2	T=0	5	m	
3	Baud rate = (clock frequency) / 372	5.7	m	
4	Bit/character duration and sampling time in accordance with ISO/IEC 7816-3	5.9	m	
5	Error detection and character repetition procedure in accordance with ISO/IEC 7816-3	5.10	m	

Comments:

A.6.2.1 Supply voltage Vcc (contact C1)

Table A.7: Supply voltage characteristics

Item	Operating characteristic	Reference	Status	Support
1	Operating voltage range 4,5 V to 5,5 V	5.1	m	
2	Current consumption ≤ 10 mA at any frequency accepted by the DAM	5.1	m	
3	Idle current consumption ≤ 200 µA at 1 MHz and 25°C	5.1	m	
4	Idle current consumption ≤ 1 mA at any frequency accepted by the DAM	5.1	m	

Comments:

A.6.2.2 Reset RST (contact C2)

Table A.8: Electrical characteristics of RST

Item	Electrical characteristic	Reference	Status	Support
1	(Vcc-0,7) ≤ V_{OH} ≤ Vcc, with I_{OLmax} = +20 µA	5.2	m	
2	0 V ≤ V_{OL} ≤ 0,6 V, with I_{OLmax} = -200 µA	5.2	m	
3	t_R t_F ≤ 400 µS, with Cout = Cin = 30 pF	5.2	m	

Comments:

A.6.2.3 Clock CLK (contact C3)

Table A.9: Electrical characteristics of CLK

Item	Electrical characteristic	Reference	Status	Support
1	1 MHz ≤ (clock frequency) ≤ 5 MHz	5.4	m	
2	Internal clock	5.4	x	
3	Duty cycle between 40 % and 60 % of the period during stable operation	5.4	m	
4	(0,7xVcc) ≤ V_{OH} ≤ Vcc, with I_{OLmax} = +20 µA	5.4	m	
5	0 V ≤ V_{OL} ≤ 0,5 V, with I_{OLmax} = -200 µA	5.4	m	
6	t_R t_F ≤ 9 % of period (0,5 µS max), with Cout = Cin = 30 pF	5.4	m	

Comments:

A.6.2.4 I/O (contact C7)

Table A.10: Electrical characteristics of I/O

Item	Electrical characteristic	Reference	Status	Support
1	$(0,7 \times Vcc) \leq V_{IH} \leq (Vcc+0,3\ V)$, $I_{IHmax} = +20\ \mu A$	5.5	m	
2	$-0,3\ V \leq V_{IL} \leq 0,8\ V$, with $I_{ILmax} = +1\ mA$	5.5	m	
3	$3,8\ V \leq V_{OH} \leq Vcc$, with $I_{OHmax} = +20\ \mu A$	5.5	m	
4	$0\ V \leq V_{OL} \leq 0,4\ V$, with $I_{OLmax} = -1\ mA$	5.5	m	
5	$t_R\ t_F \leq 1\ \mu S$, with Cout = Cin = 30 pF	5.5	m	

Comments:

A.6.2.5 States

Table A.11: Clock stop modes

Item	Clock stop mode	Reference	Status	Support
1	Clock stop allowed, no preferred level	5.6, 10.1.1, 9.2.1	o.4	
2	Clock stop allowed, high level preferred	5.6, 10.1.1, 9.2.1	o.4	
3	Clock stop allowed, low level preferred	5.6, 10.1.1, 9.2.1	o.4	
4	Clock stop not allowed	5.6, 10.1.1, 9.2.1	o.4	
5	Clock stop only allowed on high level	5.6, 10.1.1, 9.2.1	o.4	
6	Clock stop only allowed on low level	5.6, 10.1.1, 9.2.1	o.4	

o.4: It is mandatory to support exactly one of these items.

Comments:

A.6.2.6 Answer To Reset (ATR)

Table A.12: Structure, contents and PTS procedure

Item	ATR characteristic	Reference	Status	Support
1	ATR length ≤ 33 characters	5.8.1	m	
2	ATR: TS	5.8.1	m	
3	ATR: T0	5.8.1	m	
4	ATR: TA1	5.8.1	o	
5	ATR: TB1	5.8.1	o	
6	ATR: PI1 = 0	5.8.1	c1201	
7	ATR: TC1	5.8.1	o	
8	ATR: TD1	5.8.1	o	
9	ATR: TA2	5.8.1	o	
10	ATR: TB2	5.8.1	x	
11	ATR: TC2	5.8.1	o	
12	ATR: TDi (i>1)	5.8.1	o	
13	ATR: TAi, TBi, TCi (i>2)	5.8.1	o	
14	ATR: Historical characters T1,...,TK	5.8.1	o	
15	ATR: Check character	5.8.1	c1202	
16	PTS procedure	5.8.2	c1203	

c1201: IF A.12/5 THEN m ELSE n/a -- TB1 sent
c1202: IF (ATR indicates support for protocol(s) in addition to T=0) THEN m ELSE x
c1203: IF (A.12/4 AND TA1 ≠ '11') THEN m ELSE n/a -- TA1 sent AND TA1 ≠ '11'

Comments:

A.6.2.6.1 ATR: TC1 parameter values

Table A.13: ATR: TC1 parameter values

Prerequisite: A.12/7 -- TC1 sent

Item	Parameter value	Reference	Status	Support
1	ATR: TC1 = 0	5.8.1	o.5	
2	ATR: TC1 = 255	5.8.1	o.5	

o.5: It is mandatory to support exactly one of these items.

Comments:

A.6.3 Logical model

A.6.3.1 File identifier

The questions in the following table, regarding the specification of file-type IDs, apply to files in an implementation of the DECT application only. They need not be answered for files in other card applications.

Table A.14: File identifier

Item	File ID characteristic	Reference	Status	Support
1	Master file coded as '3F00'	6.2	m	
2	File type '7F' to identify Dedicated Files	6.2	m	
3	File types '00', '01' and '2F' to identify Elementary Files under the Master File	6.2	m	
4	File type '6F' to identify Elementary Files under a Dedicated File	6.2	m	
5	File ID is assigned at the time of creation of the file concerned	6.2	m	
6	Two files under the same parent never have the same ID	6.2	m	
7	A child and any parent in the direct hierarchy never have the same ID	6.2	m	

Comments:

A.6.3.2 Dedicated files

Table A.15: Dedicated files

Item	DF characteristic	Reference	Status	Support
1	DF_{DECT}	6.3	m	
2	The identifier for DF_{DECT} contained in EF_{DIR} (please specify the actual identifier below)	6.3	c1501	
3	The path to DF_{DECT} contained in EF_{DIR} (please specify the actual path below)	6.3	c1501	
4	DFs for registrations stored as immediate children under DF_{DECT}	6.3	m	
5	Each registration stored in its own DF	6.3	m	

c1501: If A.31/5 THEN m ELSE n/a -- EF_{DIR} supplied.

Comments:

A.6.3.3 Elementary files

Table A.16: File structures

Item	EF type	Reference	Status	Support
1	Transparent EF	6.4.1, 10.1.1	m	
2	Linear fixed EF	6.4.2, 10.3.1	m	
3	Cyclic EF	6.4.3	o	

Comments:

A.6.3.4 Methods for selecting the DECT application

Table A.17: Selection of the DECT application

Item	Method	Reference	Status	Support
1	DECT application selection using the file identifiers stored in EF_{DIR}	6.5	o.6	
2	Direct selection of the DECT application using the DECT application identifier	6.5	o.6	

o.6: It is mandatory to support at least one of these items.

Comments:

A.6.3.5 Reservation of file IDs

A.6.3.5.1 DFs

Table A.18: Reservation of file IDs for DFs

Item	File ID	Reference	Status	Support
1	'7F 60' for administrative use	6.7	m	
2	'7F 7X' for registrations	6.7	m	
3	'7F 50', '7F 51', '7F 52' for operational use	6.7	m	

Comments:

A.6.3.5.2 EFs

Table A.19: Reservation of file IDs for EFs

Item	File ID	Reference	Status	Support
1	'2F DX' in the MF '3F 00' for administrative use	6.7	m	
2	'6F DX' in the DFs '7F 50', '7F 51', '7F 52', '7F 7X' for administrative use	6.7	m	
3	'2F 1X' in the MF '3F 00' for operational use	6.7	m	
4	'6F XX' (except '6F DX') in the DFs '7F 50', '7F 51', '7F 52', '7F 7X' for operational use	6.7	m	

Comments:

A.6.4 Security services and facilities

The process of authentication often involves the interaction of multiple entities whereby any entity may attempt to verify the others, and provide verification of itself on request. The aim of the following questions is to determine whether the DAM implementation under test is capable of performing the necessary operations that are required of a DAM during the various authentication procedures.

A.6.4.1 Algorithms and processes

Table A.20: Algorithms and processes

Item	Algorithm/process	Reference	Status	Support
1	DECT Standard Authentication Algorithm (DSAA)	7.1.3	m	
2	Other authentication algorithms	7.1.3	o	
3	B1, as defined in ETS 300 175-7	7.1.3, 7.2.1, 7.2.2	m	
4	B2, as defined in ETS 300 175-7	7.1.3, 7.2.3	m	
5	A11, as defined in ETS 300 175-7	7.1.3, 7.2.1	m	
6	A12, as defined in ETS 300 175-7	7.1.3, 7.2.1	m	
7	A21, as defined in ETS 300 175-7	7.1.3, 7.2.2	m	
8	A22, as defined in ETS 300 175-7	7.1.3, 7.2.2	m	

Comments:

A.6.4.2 Authentication

Table A.21: Authentication mechanisms

Item	Authentication mechanism	Reference	Status	Support
1	Authentication of a Portable Termination (PT)	7.2.1	m	
2	Authentication of a Fixed Termination (FT)	7.2.2	m	
3	User authentication	7.2.3	m	
4	Direct mutual authentication	7.2.4	m	
5	Indirect mutual authentication	7.2.4	m	

Comments:

A.6.4.3 UAK allocation

Table A.22: UAK allocation mechanism

Item	Mechanism	Reference	Status	Support
1	UAK allocation	7.3	m	

Comments:

A.6.4.4 File access control

Table A.23: File access conditions

Item	Access code and condition	Reference	Status	Support
1	0: ALWAYS	7.6, 9.3	m	
2	1: CHV1	7.6, 9.3	m	
3	2: CHV2	7.6, 9.3	m	
4	3: RFU	9.3	m	
5	4: FT-AUT	7.6, 9.3	m	
6	5-7: RFU	9.3	m	
7	8: CHV1 and FT-AUT	9.3	m	
8	9: CHV2 and FT-AUT	9.3	m	
9	10-14: ADM	7.6, 9.3	m	
10	15: NEVER	7.6, 9.3	m	

Comments:

A.6.5 Description of the functions

Table A.24: Functions

Item	Function	Reference	Status	Support
1	SELECT	8.1, 6.6	m	
2	STATUS	8.2	m	
3	READ BINARY	8.3	m	
4	UPDATE BINARY	8.4	m	
5	READ RECORD	8.5	m	
6	UPDATE RECORD	8.6	m	
7	SEEK	8.7	m	
8	INCREASE	8.8	c2401	
9	VERIFY CHV	8.9	m	
10	CHANGE CHV	8.10	m	
11	DISABLE CHV	8.11	m	
12	ENABLE CHV	8.12	m	
13	UNBLOCK CHV	8.13	m	
14	INVALIDATE	8.14	m	
15	REHABILITATE	8.15	m	
16	ASK RANDOM	8.16	m	
17	PT AUTHENTICATION	8.17, 7.2.1	m	
18	FT AUTHENTICATION	8.18, 7.2.2	m	
19	USER AUTHENTICATION	8.19, 7.2.3	m	
20	UAK ALLOCATION	8.20, 7.3	m	

c2401: IF A.16/3 THEN m ELSE n/a -- Support for cyclic EFs.

Comments:

A.6.6 Description of the commands

A.6.6.1 Mapping principles

Table A.25: APDU mapping principles and parameters

Item	APDU format/parameter	Reference	Status	Support
1	Receive command with format CLA, INS, P1, P2, P3 {, data}	9.1	m	
2	Send response with format {data,} SW1, SW2	9.1	m	
3	CLA in the range A0-A3	9.1	m	

Comments:

A.6.6.2 Coding of the commands

Table A.26: Coding of the commands

Item	Command	Reference	Status	Support
1	SELECT	9.2.1	m	
2	STATUS	9.2.2	m	
3	READ BINARY	9.2.3	m	
4	UPDATE BINARY	9.2.4	m	
5	READ RECORD	9.2.5	m	
6	UPDATE RECORD	9.2.6	m	
7	SEEK	9.2.7	m	
8	INCREASE	9.2.8	c2601	
9	VERIFY CHV	9.2.9	m	
10	CHANGE CHV	9.2.10	m	
11	DISABLE CHV	9.2.11	m	
12	ENABLE CHV	9.2.12	m	
13	UNBLOCK CHV	9.2.13	m	
14	INVALIDATE	9.2.14	m	
15	REHABILITATE	9.2.15	m	
16	ASK RANDOM	9.2.16	m	
17	PT AUTHENTICATION	9.2.17	m	
18	FT AUTHENTICATION	9.2.18	m	
19	USER AUTHENTICATION	9.2.19	m	
20	UAK ALLOCATION	9.2.20	m	
21	GET RESPONSE	9.2.21	m	

c2601: IF A.24/8 THEN m ELSE n/a -- Support for INCREASE function.

Comments:

A.6.6.3 Definitions and coding

Table A.27: Coding of response parameters/data of the commands

Item	Parameter/data	Reference	Status	Support
1	RFU bytes set to '00'	9.3	c2701	
2	RFU bits set to 0	9.3	c2701	
3	File structure indicators	9.3	m	
4	File type indicators	9.3	m	

c2701: IF (DECT specific card) THEN m ELSE n/a

Comments:

A.6.6.4 Status conditions returned by the DAM

A.6.6.4.1 Coding of the status words

Table A.28: Coding of all status words by context

Item	Context of status condition	Reference	Status	Support
1	Correctly executed commands	9.4.1	m	
2	Memory management	9.4.2	m	
3	Referencing management	9.4.3	m	
4	Security management	9.4.4	m	
5	Application independent errors	9.4.5	m	

Comments:

A.6.6.4.2 Commands versus possible status responses

The questions in the following subclause concern status conditions that are returned by the card following the receipt and processing of commands. They apply to all of the specified commands, listed in table A.26.

Table A.29: Status responses to commands

Item	Status response	Reference	Status	Support
1	'90 00'	9.4.6	m	
2	'9F XX'	9.4.6	m	
3	'92 0X'	9.4.6	o	
4	'92 40'	9.4.6	m	
5	'94 00'	9.4.6	m	
6	'94 02'	9.4.6	m	
7	'94 04'	9.4.6	m	
8	'94 08'	9.4.6	m	
9	'98 02'	9.4.6	m	
10	'98 04'	9.4.6	m	
11	'98 08'	9.4.6	m	
12	'98 10'	9.4.6	m	
13	'98 35'	9.4.6	m	
14	'98 40'	9.4.6	m	
15	'98 50'	9.4.6	c2901	
16	'67 XX'	9.4.6	m	
17	'6B XX'	9.4.6	m	
18	'6D XX'	9.4.6	m	
19	'6E XX'	9.4.6	m	
20	'6F XX'	9.4.6	m	

c2901: IF A.24/8 THEN m ELSE n/a -- Support for INCREASE function.

Comments:

A.6.7 Contents of the elementary files

Table A.30: General capabilities

Item	Capability	Reference	Status	Support
1	ASCII coding in accordance with ISO 8859-1	10	m	

Comments:

A.6.7.1 Contents of the EFs at the MF level

Table A.31: EFs at the MF level

Item	EF	Reference	Status	Support
1	EF_{ICC}	10.1.1	m	
2	EF_{ID}	10.1.2	m	
3	EF_{NAME}	10.1.3	o	
4	EF_{IC}	10.1.4	o	
5	EF_{DIR}	10.1.5	o	
6	EF_{LANG}	10.1.6	o	

Comments:

A.6.7.1.1 Optional data parameters in EF_{ICC}

Table A.32: Optional data parameters in EF_{ICC}

Item	Data parameter	Reference	Status	Support
1	IC identifier	10.1.1	o	
2	Card Profile	10.1.1	o	
3	Type of selection	10.1.1	o	

Comments:

A.6.7.1.2 Optional data parameters in EF_{ID}

Table A.33: Optional data parameters in EF_{ID}

Item	Data parameter	Reference	Status	Support
1	Date of activation	10.1.2	o	
2	Card expiry date	10.1.2	o	
3	Card sequence number	10.1.2	o	
4	Country code	10.1.2	o	

Comments:

A.6.7.1.3 Optional data parameters in EF_{NAME}

Table A.34: Optional data parameters in EF_{NAME}

Item	Data parameter	Reference	Status	Support
1	Card holder name	10.1.3	o	

Comments:

A.6.7.1.4 Optional data parameters in EF_{DIR}

Table A.35: Optional data parameters in EF_{DIR}

Item	Data parameter	Reference	Status	Support
1	Application label tag	10.1.5	o	
2	Application label length	10.1.5	o	
3	Application label (verbal description)	10.1.5	o	

Comments:

A.6.7.1.5 Optional data parameters in EF_{LANG}

Table A.36: Optional data parameters in EF_{LANG}

Item	Data parameter	Reference	Status	Support
1	First language preference	10.1.6	o	
2	Second language preference	10.1.6	o	
3	Third language preference	10.1.6	o	
4	Fourth language preference	10.1.6	o	

Comments:

A.6.7.2 Contents of the EFs at the parent level of the DECT application

Table A.37: EFs at the parent level of the DECT application

Item	EF	Reference	Status	Support
1	EF_{CHV1}	10.2.1	m (note)	
2	EF_{CHV2}	10.2.1	o	
NOTE: In a mono application DAM, EF_{CHV1} shall be at the MF level.				

Comments:

A.6.7.3 Contents of the EFs at the DECT application level

Table A.38: EFs at the DECT level

Item	EF	Reference	Status	Support
1	EF_{LSR}	10.3.1	m	
2	EF_{LCSR}	10.3.2	m	
3	EF_{IPDI}	10.3.3	m	

Comments:

A.6.7.3.1 Optional data parameters in EF_{LSR}

Table A.39: Optional data parameters in EF_{LSR}

Item	Data parameter	Reference	Status	Support
1	Further subscription registrations (comment on the number of optional registrations in the implementation)	10.3.1	o	

Comments:

A.6.7.4 Contents of the EFs at the subscription registration level

A.6.7.4.1 DF_{SR1}

Table A.40: EFs at the DF_{SR1} level

Item	EF	Reference	Status	Support
1	EF_{SR}	10.4.1	m	
2	EF_{IPUI}	10.4.2	m	
3	EF_{PARK}	10.4.3	m	
4	EF_{TPUI}	10.4.4	o	
5	EF_{ZAP}	10.4.5	o	
6	EF_{DCK}	10.4.6	m	
7	EF_{UAK}	10.4.7	m	
8	EF_{AC}	10.4.8	m	
9	EF_{ST}	10.4.9	m	

Comments:

A.6.7.4.1.1 Optional data parameters in EF_{PARK}

Table A.41: Optional data parameters in EF_{PARK}

Item	Data parameter	Reference	Status	Support
1	Further PLI, ARC and ARD entries (comment on the number of optional PARK entries in the implementation)	10.4.3	o	

Comments:

A.6.7.4.1.2 Optional data parameters in EF_{UAK}

Table A.42: Optional data parameters in EF_{UAK}

Item	Data parameter	Reference	Status	Support
1	Length of UAK number 1	10.4.7	o	
2	UAK number 1	10.4.7	o	
3	Length of UAK number 2	10.4.7	o	
4	UAK number 2	10.4.7	o	
5	Length of UAK number 3	10.4.7	o	
6	UAK number 3	10.4.7	o	
7	Length of UAK number 4	10.4.7	o	
8	UAK number 4	10.4.7	o	
9	Length of UAK number 5	10.4.7	o	
10	UAK number 5	10.4.7	o	
11	Length of UAK number 6	10.4.7	o	
12	UAK number 6	10.4.7	o	
13	Length of UAK number 7	10.4.7	o	
14	UAK number 7	10.4.7	o	

Comments:

A.6.7.4.1.3 Optional data parameters in EF_{AC}

Table A.43: Optional data parameters in EF_{AC}

Item	Data parameter	Reference	Status	Support
1	Length of AC number 1	10.4.8	o	
2	AC number 1	10.4.8	o	
3	Length of AC number 2	10.4.8	o	
4	AC number 2	10.4.8	o	
5	Length of AC number 3	10.4.8	o	
6	AC number 3	10.4.8	o	
7	Length of AC number 4	10.4.8	o	
8	AC number 4	10.4.8	o	
9	Length of AC number 5	10.4.8	o	
10	AC number 5	10.4.8	o	
11	Length of AC number 6	10.4.8	o	
12	AC number 6	10.4.8	o	
13	Length of AC number 7	10.4.8	o	
14	AC number 7	10.4.8	o	

Comments:

A.6.7.4.2 DF_{SR2}

Table A.44: EFs at the DF_{SR2} level

Item	EF	Reference	Status	Support
1	EF_{SR}	10.4.1	m	
2	EF_{IPUI}	10.4.2	m	
3	EF_{PARK}	10.4.3	m	
4	EF_{TPUI}	10.4.4	o	
5	EF_{ZAP}	10.4.5	o	
6	EF_{DCK}	10.4.6	m	
7	EF_{UAK}	10.4.7	m	
8	EF_{AC}	10.4.8	m	
9	EF_{ST}	10.4.9	m	

Comments:

A.6.7.4.2.1 Optional data parameters in EF_{PARK}

Table A.45: Optional data parameters in EF_{PARK}

Item	Data parameter	Reference	Status	Support
1	Further PLI, ARC and ARD entries (comment on the number of optional PARK entries in the implementation)	10.4.3	o	

Comments:

A.6.7.4.2.2 Optional data parameters in EF_{UAK}

Table A.46: Optional data parameters in EF_{UAK}

Item	Data parameter	Reference	Status	Support
1	Length of UAK number 1	10.4.7	o	
2	UAK number 1	10.4.7	o	
3	Length of UAK number 2	10.4.7	o	
4	UAK number 2	10.4.7	o	
5	Length of UAK number 3	10.4.7	o	
6	UAK number 3	10.4.7	o	
7	Length of UAK number 4	10.4.7	o	
8	UAK number 4	10.4.7	o	
9	Length of UAK number 5	10.4.7	o	
10	UAK number 5	10.4.7	o	
11	Length of UAK number 6	10.4.7	o	
12	UAK number 6	10.4.7	o	
13	Length of UAK number 7	10.4.7	o	
14	UAK number 7	10.4.7	o	

Comments:

A.6.7.4.2.3 Optional data parameters in EF_{AC}

Table A.47: Optional data parameters in EF_{AC}

Item	Data parameter	Reference	Status	Support
1	Length of AC number 1	10.4.8	o	
2	AC number 1	10.4.8	o	
3	Length of AC number 2	10.4.8	o	
4	AC number 2	10.4.8	o	
5	Length of AC number 3	10.4.8	o	
6	AC number 3	10.4.8	o	
7	Length of AC number 4	10.4.8	o	
8	AC number 4	10.4.8	o	
9	Length of AC number 5	10.4.8	o	
10	AC number 5	10.4.8	o	
11	Length of AC number 6	10.4.8	o	
12	AC number 6	10.4.8	o	
13	Length of AC number 7	10.4.8	o	
14	AC number 7	10.4.8	o	

Comments:

A.7 DECT Portable Equipment (PE)

This clause contains the ICS proforma tables related to the PE role. They are needed to be completed only for PE implementations:

Prerequisite: A.1/2 -- PE implementation

A.7.1 Physical characteristics

Table A.48: Major capabilities

Item	Physical characteristic	Reference	Status	Support
1	ID-1 card	4.1.1	o.7	
2	Plug-in card	4.1.2	o.7	
3	Embossed ID-1 card	4.1.1	c4801	
4	Contact C4 provided	4.3.1	x	
5	Contact C6 provided	4.3.1	o	
6	Contact C8 provided	4.3.1	x	
7	Operating procedures in accordance with ISO/IEC 7816-3	4.3.2	m	
8	Deactivation of contacts in any order, provided all signals reach low level before Vcc leaves high level	4.3.2	c4802	
9	Contact activation sequence for any voltage level monitored during activation	4.3.2	m	
10	Contact deactivation sequence for any voltage level monitored during deactivation	4.3.2	m	
11	Inactive contact voltage on C1, C2, C3, C6 and C7 between 0 and ±0,4 V, referenced to ground (C5)	4.3.3	m (note)	
12	Radius of any curvature of contacting elements ≥ 0,8 mm	4.3.4	m	
13	Contact pressure ≤ 0,5N per contact	4.3.4	m	
14	ID-1 card takes precedence over plug-in card	4.4	c4803	
15	Adequate precautions taken to safeguard the PE, DAM and PE/DAM interface from static discharges at all times	4.5	m	
NOTE: When contact C6 is not provided then the reference to C6 does not apply.				

o.7: It is mandatory to support at least one of these items.
c4801: IF A.48/1 THEN m ELSE n/a -- ID-1 card supported.
c4802: IF (DAM clock is already stopped and is not restarted) THEN o ELSE n/a
c4803: IF (A.48/1 AND A.48/2) THEN m ELSE n/a -- ID-1 card and plug-in card supported.

Comments:

A.7.2 Electronic signals and transmission protocols

Table A.49: Major capabilities

Item	Capability	Reference	Status	Support
1	Electronic signals and transmission protocols in accordance with ISO/IEC 7816-3	5	m	
2	T=0	5	m	
3	Baud rate = (clock frequency) / 372	5.7	m	
4	Bit/character duration and sampling time in accordance with ISO/IEC 7816-3	5.9	m	
5	Perform a Reset on receipt of an ATR which is not in accordance to ETS 300 331 [1]	5.10	m	
6	Rejection of the DAM does not occur until at least three consecutive wrong ATRs are received	5.10	m	
7	Error detection and character repetition procedure during ATR/PTS in accordance with ISO/IEC 7816-3	5.10	o	
8	Error detection and character repetition procedure following ATR/PTS in accordance with ISO/IEC 7816-3, using T=0	5.10	m	
9	Hardware check to detect the presence of the DAM	5.11	o.8	
10	Software check to detect the presence of the DAM	5.11	o.8	

o.8: It is mandatory to support at least one of these items.

Comments:

A.7.2.1 Supply voltage Vcc (contact C1)

Table A.50: Supply voltage characteristics

Item	Operating characteristic	Reference	Status	Support
1	Supply voltage range 4,5 V to 5,5 V	5.1	m	
2	Supply current consumption ≤ 10 mA at any frequency accepted by the DAM	5.1	m	
3	Supply an idle current consumption ≤ 200 µA at 1 MHz and 25°C	5.1	m	
4	Supply an idle current consumption ≤ 1 mA at any frequency accepted by the DAM	5.1	m	
5	Counteract current consumption spikes as specified	5.1	m	

Comments:

A.7.2.2 Reset RST (contact C2)

Table A.51: Electrical characteristics of RST

Item	Electrical characteristic	Reference	Status	Support
1	(Vcc-0,7) ≤ V_{OH} ≤ Vcc, with I_{OLmax} = +20 µA	5.2	m	
2	0 V ≤ V_{OL} ≤ 0,6 V, with I_{OLmax} = -200 µA	5.2	m	
3	t_R t_F ≤ 400µS, with Cout = Cin = 30 pF	5.2	m	

Comments:

A.7.2.3 Programming voltage Vpp (contact C6)

Table A.52: Programming voltage Vpp

Item	Electrical characteristic	Reference	Status	Support
1	Vpp = Vcc, for the ID-1 card	5.3	c5201	
2	Vpp = Vcc, for the plug-in card	5.3	c5202	

c5201: IF (A.48/5 AND A.48/1) THEN m ELSE n/a -- C6 provided AND ID-1 card supported.
c5202: IF (A.48/5 AND A.48/2) THEN o ELSE n/a -- C6 provided AND plug-in card supported.

Comments:

A.7.2.4 Clock CLK (contact C3)

Table A.53: Electrical characteristics of CLK

Item	Electrical characteristic	Reference	Status	Support
1	1 MHz ≤ (clock frequency) ≤ 5 MHz	5.4	m	
2	Duty cycle between 40 % and 60 % of the period during stable operation	5.4	m	
3	(0,7xVcc) ≤ V_{OH} ≤ Vcc, with I_{OLmax} = +20 µA	5.4	m	
4	0 V ≤ V_{OL} ≤ 0,5 V, with I_{OLmax} = -200 µA	5.4	m	
5	t_R t_F ≤ 9 % of period (0,5µS max), with Cout = Cin = 30 pF	5.4	m	

Comments:

A.7.2.5 I/O (contact C7)

Table A.54: Electrical characteristics of I/O

Item	Electrical characteristic	Reference	Status	Support
1	$(0,7 \times Vcc) \leq V_{IH} \leq (Vcc+0,3\ V)$, $I_{IHmax} = +20\ \mu A$	5.5	m	
2	$-0,3\ V \leq V_{IL} \leq 0,8\ V$, with $I_{ILmax} = +1\ mA$	5.5	m	
3	$3,8\ V \leq V_{OH} \leq Vcc$, with $I_{OHmax} = +20\ \mu A$	5.5	m	
4	$0\ V \leq V_{OL} \leq 0,4\ V$, with $I_{OLmax} = -1\ mA$	5.5	m	
5	$t_R\ t_F \leq 1 \mu S$, with Cout = Cin = 30 pF	5.5	m	

Comments:

A.7.2.6 States

Table A.55: Clock stop operation

Item	Clock stop operation	Reference	Status	Support
1	Clock stop	5.6	m	

Comments:

A.7.2.7 Answer To Reset (ATR)

Table A.56: Structure, contents and PTS procedure

Item	ATR characteristic	Reference	Status	Support
1	ATR length ≤ 33 characters	5.8.1	m	
2	ATR: TS evaluation	5.8.1	m	
3	ATR: T0 evaluation	5.8.1	m	
4	ATR: TA1 evaluation	5.8.1	m	
5	ATR: TB1 evaluation	5.8.1	m	
6	ATR: TC1 evaluation	5.8.1	m	
7	ATR: TD1 evaluation	5.8.1	m	
8	ATR: TA2 evaluation	5.8.1	o	
9	ATR: TB2 evaluation	5.8.1	n/a	
10	ATR: TC2 evaluation	5.8.1	m	
11	ATR: TDi (i>1) evaluation	5.8.1	m	
12	ATR: TAi, TBi, TCi (i>2) evaluation	5.8.1	o	
13	ATR: Historical characters, T1,...,TK evaluation	5.8.1	o	
14	ATR: Check character evaluation	5.8.1	o	
15	PTS procedure	5.8.2	m	

Comments:

A.7.3 Logical model

A.7.3.1 Methods for selecting the DECT application

Table A.57: Selecting the DECT application

Item	Method	Reference	Status	Support
1	DECT application selection using the file identifiers stored in EF_{DIR}	6.5	m	
2	Direct selection of the DECT application using the DECT application identifier	6.5	m	

Comments:

A.7.4 Description of the commands

A.7.4.1 Mapping principles

Table A.58: APDU mapping principles

Item	APDU parameter	Reference	Status	Support
1	Send command with format CLA, INS, P1, P2, P3 {, data}	9.1	m	
2	Receive response with format {data,} SW1, SW2	9.1	m	
3	CLA = A0 as default	9.1	m	

Comments:

A.7.4.2 Coding of the commands

Table A.59: Coding of the commands

Item	Command	Reference	Status	Support
1	SELECT	9.2.1	m	
2	STATUS	9.2.2	m	
3	READ BINARY	9.2.3	m	
4	UPDATE BINARY	9.2.4	m	
5	READ RECORD	9.2.5	m	
6	UPDATE RECORD	9.2.6	m	
7	SEEK	9.2.7	m	
8	INCREASE	9.2.8	o	
9	VERIFY CHV	9.2.9	m	
10	CHANGE CHV	9.2.10	m	
11	DISABLE CHV	9.2.11	m	
12	ENABLE CHV	9.2.12	m	
13	UNBLOCK CHV	9.2.13	m	
14	INVALIDATE	9.2.14	m	
15	REHABILITATE	9.2.15	m	
16	ASK RANDOM	9.2.16	m	
17	PT AUTHENTICATION	9.2.17	m	
18	FT AUTHENTICATION	9.2.18	m	
19	USER AUTHENTICATION	9.2.19	m	
20	UAK ALLOCATION	9.2.20	m	
21	GET RESPONSE	9.2.21	m	

Comments:

A.7.5 Application protocol

A.7.5.1 General procedures

Table A.60: General procedures

Item	Procedure	Reference	Status	Support
1	Reading an EF	11.1.1	m	
2	Updating an EF	11.1.2	m	
3	Increasing an EF	11.1.3	o	

Comments:

A.7.5.2 DAM management procedures

Table A.61: DAM management procedures

Item	Procedure	Reference	Status	Support
1	DAM initialization	11.2.1	m	
2	DAM session termination	11.2.2	m	
3	Language preference	11.2.3	m	
4	Service table request	11.2.4	m	
5	DAM presence detection	11.2.5	m	

Comments:

A.7.5.3 CHV related procedures

Table A.62: CHV related procedures

Item	Procedure	Reference	Status	Support
1	CHV verification	11.3.1	m	
2	CHV value substitution	11.3.2	m	
3	CHV disabling	11.3.3	m	
4	CHV enabling	11.3.4	m	
5	CHV unblocking	11.3.5	m	

Comments:

A.7.5.4 Authentication procedures

Table A.63: Authentication procedures

Item	Procedure	Reference	Status	Support
1	Authentication of a PT	11.4.1	m	
2	Authentication of an FT	11.4.2	m	
3	User authentication	11.4.3	m	
4	Mutual authentication	11.4.4	m	

Comments:

A.7.5.5 UAK allocation

Table A.64: UAK allocation procedure

Item	Procedure	Reference	Status	Support
1	UAK allocation	11.5	m	

Comments:

A.7.5.6 General information procedures

Table A.65: General information procedures

Item	Procedure	Reference	Status	Support
1	EF_{ICC} request	11.6.1	o	
2	EF_{ID} request	11.6.2	o	
3	EF_{NAME} request	11.6.3	o	
4	EF_{IC} request	11.6.4	o	

Comments:

A.7.5.7 Subscription registration maintenance

Table A.66: Subscription registration procedures

Item	Procedure	Reference	Status	Support
1	Entering a new subscription registration	11.7.1	m	
2	Updating an existing subscription registration	11.7.2	m	
3	Terminating an existing subscription registration	11.7.3	m	

Comments:

Annex B (informative): Bibliography

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- ISO 8859-1 (1987): "Information processing - 8-bit single byte coded graphic character sets, Part 1: Latin alphabet No.1".

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- ISO 7811-3 (1985): "Identification cards - Recording technique - Part 3: Location of embossed character on ID-1 cards".

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